

Colloït-Théline-Voisin counterexample to the integral Hodge conjecture

Charlie Wu

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Abstract

These are talk notes for my talk for Elden's seminar on unramified cohomology. We will be talking about the counterexample to the integral Hodge conjecture found by Colloït-Théline and Voisin.

1 Introduction

Let X be a smooth projective connected variety over $\mathbb{C}$ of dimension n , and let $Z \in \text{CH}^j(X)$ be an algebraic cycle of codimension j . By Poincaré duality, we can associate a cohomology class $[Z] \in H^{2j}(X, \mathbb{Z}(j))$. The map $i : \text{CH}^j(X) \rightarrow H^{2j}(X, \mathbb{Z}(j))$ given by $Z \mapsto [Z]$ is called the cycle class map. A computation tells us that the image of the cycle class map lies in $\text{Hdg}^{2j}(X, \mathbb{Z}) := H^{j,j}(X, \mathbb{C}) \cap H^{2k}(X, \mathbb{Z}(j))$.

We recall the integral Hodge conjecture.

Conjecture 1.1 - Integral Hodge conjecture. Let X be a smooth projective connected variety over $\mathbb{C}$ of dimension d . Then, the map $\text{CH}^j(X) \rightarrow \text{Hdg}^{2j}(X, \mathbb{Z})$ is surjective.

This conjecture is false (this is the point of the whole talk!), but there are some cases where it is true.

Proposition 1.2 - Lefschetz-(1,1) theorem. Let X be a smooth projective connected variety over $\mathbb{C}$. Then, the map $\text{CH}^1(X) \rightarrow \text{Hdg}^2(X, \mathbb{Z})$ is surjective.

Proof. We consider the exponential exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^\times \rightarrow 0.$$

This gives us a long exact sequence on cohomology, and upon identifying $H^1(X, \mathcal{O}_X^\times)$ with $\text{Pic}(X)$ and $H^2(X, \mathcal{O}_X)$ with $H^{0,2}(X, \mathbb{Z})$ we have the following exact sequence

$$\cdots \rightarrow H^1(X, \mathcal{O}_X) \rightarrow \text{Pic}(X) \rightarrow H^2(X, \mathbb{Z}) \rightarrow H^{0,2}(X, \mathbb{Z}) \rightarrow \cdots$$

Here, we are identifying the first Chern class $c_1 : \text{Pic}(X) \rightarrow H^2(X, \mathbb{Z})$ with the cycle class map. But it maps $\mathcal{O}_X(D)$ to $[D]$ so this is okay. Similarly, $H^2(X, \mathbb{Z}) \rightarrow H^{0,2}(X, \mathbb{Z})$ is given by projecting onto the $(0, 2)$ part of the Hodge decomposition. By exactness, $\ker(H^2(X, \mathbb{Z}) \rightarrow H^{0,2}(X, \mathbb{Z})) = \text{Im}(c_1)$ and this contains $H^{1,1}(X, \mathbb{Z})$ so we are done. $\square$

So, the integral Hodge conjecture is true for curves as well as any variety whose Hodge classes are generated by divisors.

2 Obstructions

For a given j , we set $Z^{2j}(X) := \text{Hdg}^{2j}(X, \mathbb{Z}) / \text{Im}(i)$ where $i : \text{CH}^j(X) \rightarrow \text{Hdg}^{2j}(X, \mathbb{Z})$ is the cycle class map. Clearly the integral Hodge conjecture is true for cycles of codimension j if and only if $Z^{2j}(X) = 0$.

The CTV strategy is to compare $Z^{2j}(X)$ to unramified cohomology, and then to find an example where unramified cohomology does not vanish.

We set $\mathcal{H}_X^i(A)$ to be the sheaf given by $U \mapsto H^i(U(\mathbb{C}), A)$ (where A is an abelian group), and we set $H_{nr}^i(X, A) = H^0(X, \mathcal{H}_X^i(A))$.

The main theorem (at least for our purposes) of Colloït-Théline-Voisin is the following:

Theorem CTV Theorem 3.9. Let X be a smooth projective variety over $\mathbb{C}$ of degree n , and suppose CH^n is supported on a variety of codimension at most 3 (think this means there is a variety Y (of dimension at least $\dim X - 3$ and a morphism $f : Y \rightarrow X$ so that $f_* : \mathrm{CH}_0(Y) \rightarrow \mathrm{CH}_0(X)$ is surjective) . Then we have an isomorphism of finite groups

$$H_{nr}^3(X, \mathbb{Q}/\mathbb{Z}(2)) \cong Z^4(X)$$

and $H_{nr}^3(X, \mathbb{Q}/\mathbb{Z}(2))$ is obtained as a union of subgroups $H_{nr}^3(X, \mu_n^{\otimes 2})$.

The counterexample is a projective connected smooth variety X over $\mathbb{C}$ of dimension 6 is unirational and $Z^4(X) \neq 0$. (This is especially interesting, since we have a counterexample immediately once we leave the world of 1-cocycles. Hence, the Lefschetz-(1, 1) theorem is actually the strongest possible statement we can make.)

3 Why CTV Theorem 3.9 is true

What we are actually going to show is that there is an exact sequence

$$0 \rightarrow H_{nr}^3(X, \mathbb{Z}(2)) \otimes \mathbb{Q}/\mathbb{Z} \rightarrow H_{nr}^3(X, \mathbb{Q}/\mathbb{Z}(2)) \rightarrow Z^4(X)_{\mathrm{tors}} \rightarrow 0.$$

To do so, we show there is an exact sequence

$$0 \rightarrow H_{nr}^3(X, \mathbb{Z}(2))/n \rightarrow H_{nr}^3(X, \mu_n^{\otimes 2}) \rightarrow Z^4(X)[n] \rightarrow 0.$$

Then, the above exact sequence is obtained by passing through limits.

Here are the rough steps needed to go into the proof.

The following computation is really the main theorem of the talk. This is the heart of Theorem 3.9 in CTV.

Lemma CTV Theorem 3.1. This sequence

$$0 \rightarrow \mathcal{H}_X^p(\mathbb{Z}(i)) \rightarrow \mathcal{H}_X^p(\mathbb{Z}(i)) \rightarrow \mathcal{H}_X^p(\mu_n^{\otimes i}) \rightarrow 0$$

is exact. This implies that $\mathcal{H}_X^p(\mathbb{Z}(i))$ (and in particular $H^0(X, \mathcal{H}_X^p(\mathbb{Z}(i)))$) is torsion free since $\mathcal{H}_X^p(\mathbb{Z}(i)) \rightarrow \mathcal{H}_X^p(\mu_n^{\otimes i})$ is never an isomorphism.

Proof. Let X_{cl} be the site of (analytic) local isomorphisms on $X(\mathbb{C})$. Let $f : X_{cl} \rightarrow X_{et}$ and $g : X_{et} \rightarrow X_{zar}$ be morphisms of sites, with $g \circ f = \pi : X_{cl} \rightarrow X_{zar}$ the standard morphism of sites from the analytic site to the Zariski site.

We have a short exact sequence

$$0 \rightarrow \mathbb{Z}(i) \rightarrow \mathbb{Z}(i) \rightarrow \mu_n^{\otimes i} \rightarrow 0$$

given by $\times n : \mathbb{Z}(i) \rightarrow \mathbb{Z}(i)$.

This gives us a long exact sequence

$$\dots \mathcal{H}_X^{p-1}(\mu_n^{\otimes i}) \rightarrow \mathcal{H}_X^p(\mathbb{Z}(i)) \rightarrow \mathcal{H}_X^p(\mathbb{Z}(i)) \rightarrow \mathcal{H}_X^p(\mu_n^{\otimes i}) \rightarrow \mathcal{H}_X^{p+1}(\mathbb{Z}(i)) \rightarrow \dots$$

To get exactness, we just need to show that $\mathcal{H}_X^p(\mathbb{Z}(i)) \rightarrow \mathcal{H}_X^p(\mu_n^{\otimes i})$ is surjective, or that $\mathcal{H}_X^p(\mathbb{Z}(i)) \rightarrow \mathcal{H}_X^p(\mathbb{Z}(i))$ is injective.

To show surjectivity, it is enough to show that $\mathcal{H}_X^p(\mathbb{Z}(p)) \rightarrow \mathcal{H}_X^p(\mu_n^{\otimes p})$ is surjective (because abstractly, $\mathbb{Z}(i) \cong \mathbb{Z}$ and $\mu_n^{\otimes i} \cong \mu_n$ over $\mathbb{C}$. That is, we can forget the twists if we want.). We first do this for $p = 1$.

We begin with the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z}(1) & \longrightarrow & \mathcal{O}_{X_{cl}} & \xrightarrow{\exp} & \mathcal{O}_{X_{cl}}^\times \longrightarrow 0 \\ & & \downarrow z \mapsto \exp(z/n) & & \downarrow z \mapsto \exp(z/n) & & \downarrow \mathrm{id} \\ 0 & \longrightarrow & \mu_n & \longrightarrow & \mathcal{O}_{X_{cl}} & \xrightarrow{z \mapsto z^n} & \mathcal{O}_{X_{cl}}^\times \longrightarrow 0 \end{array} .$$

Here, X_{cl} is the site of local isomorphisms (on the analytic topology). We then get a diagram

$$\begin{array}{ccc} \mathcal{O}_{X_{cl}}^\times & \longrightarrow & R^1\pi_*\mathbb{Z}(1) \\ \downarrow & & \downarrow \\ \mathcal{O}_{X_{cl}}^\times & \longrightarrow & R^1\pi_*\mu_n \end{array}.$$

where $\pi : X_{cl} \rightarrow X_{zar}$ is a morphism of sites. Then, we get a morphism of sheaves on the Zariski site

$$\mathcal{O}_{X_{zar}}^\times \rightarrow \pi_*\mathcal{O}_{X_{cl}}^\times \rightarrow R^1\pi_*\mu_n.$$

Similarly, we have a morphism $f : X_{cl} \rightarrow X_{et}$ of sites which gives us an exact sequence

$$1 \rightarrow f_*\mu_n \rightarrow f_*\mathcal{O}_{X_{et}} \rightarrow f_*\mathcal{O}_{X_{et}} \rightarrow R^1f_*\mu_n$$

and by a theorem of Artin, we have that $R^qf_*F = 0$ for all $q > 0$ whenever F is a locally constant torsion sheaf on X_{cl} .

Then, we get the following commutative diagram **What are the vertical maps? I think they are given by $\mathcal{O}_{X_{et}}^\times \rightarrow f_*\mathcal{O}_{X_{cl}}^\times$ and then restricting the action to the invertible elements**

$$\begin{array}{ccccccc} 1 & \longrightarrow & f_*\mu_n & \longrightarrow & f_*\mathcal{O}_{X_{cl}}^\times & \xrightarrow{\times n} & f_*\mathcal{O}_{X_{cl}}^\times \longrightarrow 1 \\ & & \uparrow \sim & & \uparrow & & \uparrow \\ 1 & \longrightarrow & \mu_n & \longrightarrow & \mathcal{O}_{X_{et}}^\times & \xrightarrow{\times n} & \mathcal{O}_{X_{et}}^\times \longrightarrow 1 \end{array}$$

Pushing forward along g to the étale site and truncating on the left, we get the diagram

$$\begin{array}{ccc} \pi_*\mathcal{O}_{X_{cl}}^\times & \xrightarrow{\sim} & g_*f_*\mathcal{O}_{X_{cl}}^\times \longrightarrow R^1g_*(f_*\mu_n) \\ & & \downarrow \quad \quad \downarrow \sim \\ & & g_*\mathcal{O}_{X_{et}}^\times \longrightarrow R^1g_*\mu_n \end{array}.$$

The Grothendieck spectral sequence tells us that there is a spectral sequence so that on the E_2 -page, $E_2^{p,q} = R^qg_*(R^pf_*\mu_n)$ converging to $R^{p+q}(f \circ g)_*\mu_n$. That is, the $E_2^{p,q}$ are the associated graded of $R^{p+q}(f \circ g)_*\mu_n$. But since $R^1f_*\mu_n = 0$ we know that $g_*(R^1f_*\mu_n) = 0$ and so $R^1\pi_*\mu_n$ is isomorphic to $R^1g_*(f_*\mu_n)$. Then, we get the diagram

$$\begin{array}{ccc} \pi_*\mathcal{O}_{X_{cl}}^\times & \longrightarrow & R^1\pi_*\mathbb{Z}(1) \\ \downarrow & & \downarrow \\ \pi_*\mathcal{O}_{X_{cl}}^\times & \longrightarrow & R^1\pi_*\mu_n \\ \uparrow & & \uparrow \sim \\ g_*\mathcal{O}_{X_{et}}^\times & \longrightarrow & R^1g_*\mu_n \end{array}.$$

Since $g_*\mathcal{O}_{X_{cl}}^\times \rightarrow R^1g_*\mu_n$ is surjective since $R^1g_*\mathcal{O}_{X_{cl}}^\times = 0$ by Hilbert's Theorem 90, this implies that $R^1\pi_*\mathbb{Z}(1) \rightarrow R^1\pi_*\mu_n$ is surjective. This is exactly what we wanted to show.

Now, we deal tensor products and cup products. To do so, we need the input of Bloch-Kato.

We get a map $\mathcal{O}_{X_{zar}}^\times \rightarrow R^1\pi_*(\mathbb{Z}(1)) \rightarrow R^1\pi_*(\mu_n) \cong R^1g_*(\mu_n)$. We know that $R^1\pi_*(\mathbb{Z}(1)) \rightarrow R^1\pi_*(\mu_n)$ is surjective by the above argument. Then, the cup product gives us a map

$$(\mathcal{O}_{X_{zar}}^\times)^{\otimes p} \rightarrow R^p\pi_*(\mathbb{Z}(p)) \rightarrow R^pg_*(\mu_n^{\otimes p}).$$

Here, we use the theorem of Artin again and the Grothendieck spectral sequence to show that $R^p\pi_*(\mu_n^{\otimes p}) \cong R^pg_*(\mu_n^{\otimes p})$. Bloch-Kato tells us that the composition $(\mathcal{O}_{X_{zar}}^\times)^{\otimes p} \rightarrow R^pg_*(\mu_n^{\otimes p})$ is surjective, so we get that $R^p\pi_*(\mathbb{Z}(p)) \rightarrow R^pg_*(\mu_n^{\otimes p})$ is surjective as desired.

The reason Bloch-Kato tells us that $(\mathcal{O}_{X_{zar}}^\times)^{\otimes p} \rightarrow R^pg_*(\mu_n^{\otimes p})$ is surjective is because $K^1/n \cong \mathcal{H}^1(\mu_n)$ and in the following diagram (where the maps are given by cup product) **maybe put $\smile^p \mathcal{H}_X^1(\mu_n)$ and $\smile^p K_1/n$ to mess with people**

$$\begin{array}{ccc} \wedge^p \mathcal{H}_X^1(\mu_n) & \longrightarrow & \mathcal{H}_X^p(\mu_n^{\otimes p}) \\ \uparrow \cong & & \uparrow \cong \\ \wedge^p K_1/n & \xrightarrow[\text{surj. by definition}]{} & K_p/n \end{array},$$

and Bloch-Kato tells us that $K_p/n \rightarrow \mathcal{H}_X^p(\mu_n^{\otimes p})$ is an isomorphism. Furthermore, $K_1/n \cong \mathcal{O}_X^\times/n$ so we are done. $\square$

Proposition CTV Proposition 3.3(i). Let X be a smooth projective connected variety over $\mathbb{C}$ of dimension d . If there is some $Y \subseteq X$ so that $\mathrm{CH}_0(Y) \rightarrow \mathrm{CH}_0(X)$ is surjective and $\dim Y = r$ (for $r < d$), then $H^0(X, \mathcal{H}_X^p(\mathbb{Z})) = 0$ for $p > r$.

Proof. Let $p > r$.

A theorem of Bloch-Srinivas says that there is some integer $N > 0$ and classes Γ_1 and Γ_2 in $\mathrm{CH}^d(X \times X)$ such that $N\Delta_X = \Gamma_1 + \Gamma_2$ and Γ_1 is supported in $Y \times X$ and Γ_2 is supported in $X \times D$ where D is a divisor.

Since X is separated, we can pull back along the diagonal embedding to get that

$$N \mathrm{id} = \Gamma_1 * + \Gamma_2 * : H^0(X, \mathcal{H}_X^p(A)) \rightarrow H^0(X, \mathcal{H}_X^p(A)).$$

After resolving singularities for Y and D if necessary, we get that the map $\Gamma_1 *$ factors through the restriction $H^0(X, \mathcal{H}_X^p(A)) \rightarrow H^0(Y, \mathcal{H}_Y^p(A))$. But $\mathcal{H}_Y^p(A) = 0$ since $\dim Y = r < p$.

By a theorem of Bloch-Ogus, So, $\Gamma_2 * = 0$. Therefore, $N \mathrm{id} = 0$ so $H^0(X, \mathcal{H}_X^p(\mathbb{Z}))$ is N -torsion. But $H^0(X, \mathcal{H}_X^p(\mathbb{Z}))$ is torsion free, so $H^0(X, \mathcal{H}_X^p(\mathbb{Z})) = 0$. $\square$

Proof of CTV Theorem 3.7. There is a spectral sequence with E_2 -page given by $E_2^{pq} = H^p(X, \mathcal{H}_X^q(\mathbb{Z}(2)))$ converging to $H^\bullet(X(\mathbb{C}), \mathbb{Z}(2))$, and this is concentrated in the half-quadrant $p \leq q$.

From the spectral sequence, we get the following exact sequence

$$0 \rightarrow H^1(X, \mathcal{H}_X^3(\mathbb{Z}(2))) \rightarrow [H^4(X(\mathbb{C}), \mathbb{Z}(2))/H_{\mathrm{alg}}^4(X(\mathbb{C}), \mathbb{Z}(2))] \rightarrow H^0(X, \mathcal{H}_X^4(\mathbb{Z}(2))).$$

We know by definition that $Z^4(X) = H^4(X(\mathbb{C}), \mathbb{Z}(2))/H_{\mathrm{alg}}^4(X(\mathbb{C}), \mathbb{Z}(2))$. Since $H^0(X, \mathcal{H}_X^4(\mathbb{Z}(2)))$ is torsion free, we conclude that $Z^4(X)[n] \cong H^1(X, \mathcal{H}_X^3(\mathbb{Z}(2)))[n]$.

Then, recall that

$$0 \longrightarrow \mathcal{H}_X^3(\mathbb{Z}(2)) \xrightarrow{\times n} \mathcal{H}_X^3(\mathbb{Z}(2)) \longrightarrow \mathcal{H}_X^3(\mu_n^{\otimes 2}) \longrightarrow 0$$

is exact, and hence we get the following exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(X, \mathcal{H}_X^3(\mathbb{Z}(2))) & \xrightarrow{\times n} & H^0(X, \mathcal{H}_X^3(\mathbb{Z}(2))) & \longrightarrow & H^0(X, \mathcal{H}_X^3(\mu_n^{\otimes 2})) \longrightarrow H^1(X, \mathcal{H}_X^3(\mathbb{Z}(2))) \\ & & & & & \nearrow \times n & \\ & & H^1(X, \mathcal{H}_X^3(\mathbb{Z}(2))) & \xleftarrow{\quad} & \dots & & \end{array}$$

Then by quotienting out by the image of the $\times n$ map on the left and taking n -torsion on the right, (i.e., taking the kernel of the $\times n$ map on H^1) we get the exact sequence

$$0 \rightarrow H^0(X, \mathcal{H}_X^3(\mathbb{Z}(2)))/n \rightarrow H^0(X, \mathcal{H}_X^3(\mu_n^{\otimes 2})) \rightarrow H^1(X, \mathcal{H}_X^3(\mathbb{Z}(2)))[n] \rightarrow 0.$$

Then rewriting, and using that $Z^4(X)[n] \cong H^1(X, \mathcal{H}_X^3(\mathbb{Z}(2)))[n]$, we get that

$$0 \rightarrow H_{nr}^3(X, \mathbb{Z}(2))/n \rightarrow H_{nr}^3(X, \mu_n^{\otimes 2}) \rightarrow Z^4(X)[n] \rightarrow 0$$

is exact, as desired. $\square$

Theorem CTV - 3.9. If there is a map $f : Y \rightarrow X$ such that $\mathrm{codim}_X Y \leq 3$ and $f_* : \mathrm{CH}_0(Y) \rightarrow \mathrm{CH}_0(X)$ is surjective, then $H_{nr}^3(X, \mathbb{Z}(2))$ is zero and so we get an isomorphism $H_{nr}^3(X, \mathbb{Q}/\mathbb{Z}(2)) \cong Z^4(X)_{\mathrm{tors}}$.

Proof. By CTV Theorem 3.7, we can just show that $H_{nr}^3(X, \mathbb{Z}(2))$ is zero. But this is the content of (Proposition 3.3(i) in CTV). $\square$

4 The counterexample

As seen in sections 2 and 3, we just need to find some X so that $H_{nr}^3(X, \mu_n^{\otimes 2})$ is nonzero, and X satisfies the condition that there is some Y of dimension at most 3 and a map $f : Y \rightarrow X$ which is surjective on CH_0 .

The answer is yes!

Theorem CTV - 5.6 (Colliot-Thélène-Ojanguren). There exists a smooth projective connected unirational variety X of dimension 6 such that $H_{nr}(X, \mu_2) \neq 0$.

Proof. Given some $f_1, f_2, g_1, g_2 \in L = \mathbb{C}(x, y, z)$, consider the quadric hypersurface in $\mathbb{P}^4$ (with homogeneous coordinates $t_0, \dots, t_4$) given by

$$t_0^2 + f_1 t_1^2 + f_2 t_2^2 + f_1 f_2 t_3^2 + g_1 g_2 t_4^2 = 0.$$

Then, this gives us a family of quadric 3-folds over $\mathbb{P}^3$. Let X be this family. Then, we get an extension of fields K/L where $K = k(X)$.

Let $Q \rightarrow X$ be the inclusion of a fiber. We claim that $\text{CH}_0(Q) \rightarrow \text{CH}_0(X)$ is an isomorphism. This is because any other point in $\text{CH}_0(X)$ is rationally equivalent to some point in Q since we can consider some open $U \subseteq \mathbb{P}^3$ upon which $X|_U \cong U \times_k Q$. In this case, it is obvious that $\text{CH}_0(Q) \rightarrow \text{CH}_0(X)$ is an isomorphism.

We give a sketch that $H_{nr}^3(X, \mu_2)$ is nonzero. We can express the unramified cohomology groups in the following form:

$$H_{nr}^3(X, \mu_2) = \ker (H^3(k(X), \mu_2) \rightarrow \bigoplus_v H^2(k(v), \mathbb{Z}/2))$$

where the direct sum runs over all valuations, and the cohomology is Galois cohomology.

We consider the image of $(f_1, f_2, g_1) \in H^3(L, \mu_2^{\otimes 3})$ under the map $H^3(L, \mu_2^{\otimes 3}) \rightarrow H^3(K, \mu_2^{\otimes 3})$.

We claim that the image is an unramified element and is nonzero.

We will black-box the fact that (f_1, f_2, f_3) is an unramified element of $H^3(L, \mu_2^{\otimes 3})$.

To show that it is nonzero, we cite a theorem of Arason which says the kernel of the map $H^3(L, \mu_2^{\otimes 3}) \rightarrow H^3(K, \mu_2^{\otimes 3})$ is generated by the cup-product (how is this a cup product?) $(f_1, f_2, g_1 g_2)$.

So either in $H^3(L, \mu_2^{\otimes 3})$, we have that $(f_1, f_2, g_1) = 0$ or $(f_1, f_2, g_1) = (f_1, f_2, g_1 g_2)$. But neither of these are possible due to a theorem of Colloit-Thelene and Ojanguren.

To show that it is unramified, we let B be a DVR with fraction field K . If B contains L , then (f_1, f_2, g_1) vanishes on $H^2(B, \mu_2^{\otimes 3})$. Otherwise, we set $A = B \cap L$. Then, we have a diagram

$$\begin{array}{ccc} H^3(K, \mu_2^{\otimes 3}) & \longrightarrow & H^2(k_B, \mu_2^{\otimes 2}) \\ \uparrow & & \uparrow \\ H^3(L, \mu_2^{\otimes 3}) & \longrightarrow & H^2(k_A, \mu_2^{\otimes 2}) \end{array}.$$

But we know that a DVR for L will kill (f_1, f_2, g_1) since (f_1, f_2, g_1) is unramified, so we are done.

It's very much not clear to me how to do the first step. CT-O argues by saying that

□